0/1 Knapsack - Dynamic Programming

Ques-) Weights = {3, 4, 6, 5}
Profits = {2, 3, 1, 4}

$\left[\begin{array}{l} W = 8 \\ n = 4 \end{array}\right]$

Ans

$$\boxed{m[i,w] = \max\left(m[i-1,w],\ m[i-1, w-w[i]] + P[i]\right)}$$

Enter Profit values in box

Arrange in Asc order

P_i	W_i
2	3
3	4
4	5
1	6

i \ weight	0	1	2	3	4	5	6	7	8
0	0	0	0	0	0	0	0	0	0
1	0	0	0	2 (3+1)	2 (4-3=1 check)	2	2	2	2
2	0	0	0	2	3	3	3	(3+2, 2) 5	(3+2, 2) 5
3	0	0	0	2	3	4	4	(4+0, 5) 5	(4+2, 5) 6
4	0	0	0	2	3	4	4 (1+4)=4	(4+0, 5) 5	(4+0, 6) 6

4-3 = 1 → check 1 wt. above row

max (2+0, 0) → 2 (above)

so max is $\boxed{2}$

check in up row we have 2 then paste it

we selected 6 so 6-4=2 check (2) in this row above not same so

check if above is same no. if same then leave and paste above

$x = \{1\ 0\ 0\ 1\}$ (3, 4, 6, 5)

6-4=2 } 3rd row
2-2=0 } 1 row

OR

$x = \{1\ 0\ 0\ 1\}$ (3, 4, 6, 5)

4th row there is 6 max check for above if same move pointer above and update 4th = 0 [6 weight] yes there is 6 check again above no 6 above update 1 [3rd row, 5 = weight] so we selected 6 in 3rd row at 8th column

8 − 5 = 3 (weight) so move to 3rd column in above row (2nd row) there is 2 then check if there is 2 in above or yes there is 2 so update 2nd row [4 = weight] = 0 and select 1st row update 1

Practice Set

Items	Apple	Orange	Banana	Melon
W	2	3	1	4
P	4	5	3	7

Capacity = 5

n = 4

P	W	i \ wt	0	1	2	3	4	5	
		0	0	0	0	0	0	0	
3	1	1	0	3 ←	(1+0, 0) 3	(3+0, 0) 3	(3+0, 0) 3	3	↑ 3 present
4	2	2	0	3	(4, 3) 4	(4+3, 3) 7	(4+3, 3) 7	7	↑ 3 present
5	3	3	0	3 ←	4	(5, 7) 7	(5+3, 7) 8	(5+4, 7) 9	↑ yes 3 pres
7	4	4	0	3	4	7	(7, 8) 8	(7+3, 9) 10	←

$x = \{ \underset{2}{0} \ \underset{3}{0} \ \underset{1}{1} \ \underset{4}{1} \}$ $\quad 10 - 7 = \boxed{3}$

Answer